

GES-Tr: Gesture Translator(Indian Context)

To Translate Indian Sign Language into Readable Sentences

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Abstract—Many of us have met or at least seen people use sign language. It is used by differently-abled people with some speech disorders to communicate with others. Sign language uses gestures made using the hand that has their generic meaning in their respective languages. Thus Sign language is different for different countries and different languages. Fig 1 shows the Indian sign language that includes 26 alphabets.

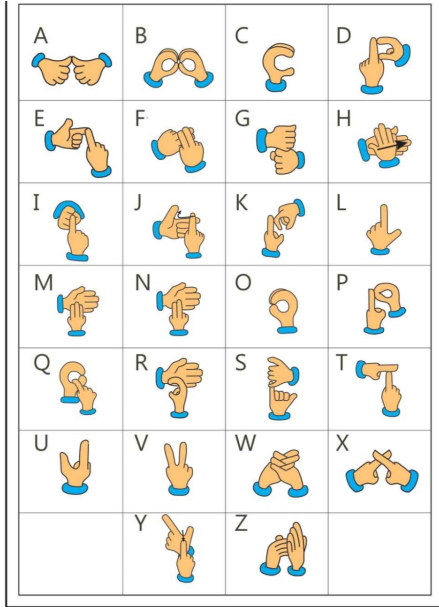


Fig. 1: Indian Sign Language

Our aim with this project is to make a device that senses hand movements and gestures to translate the Indian Sign Language into Readable Sentences so that:

- We can raise awareness about the ingenuity of Indian Sign Language and
- Create a system that can translate sign language gestures into text which can be understood by people who are not fluent in sign language.
- Minimize the role of a translator and the chances of mis-interpretation in a conversation involving a person using Sign Language and another using verbal communication.

Our device consists of two Gloves connected with sensors and an LED screen displaying the translated output. Past attempts at this project employing Computer Vision were inefficient, as the sentences did not translate entirely and were non-portable. We hope to remedy these shortcomings with the methods used in our project. This device can further be used to translate

American Sign Language(ASL), Japanese Sign Language(JSL), and many more by using the correct training dataset.

I. INTRODUCTION

Our project aims to create a device that can translate sign language into spoken or written language to aid communication for the deaf and dumb community. The device will use flexible sensors to detect the movement and positions of the user's hand and fingers as they sign and then convert those movements into spoken or written language using machine learning algorithms.

This technology has the potential to greatly enhance communication for individuals who use sign language as their primary means of communication and provide a new level of accessibility and inclusivity. Our project aims to create a cost-effective, easy-to-use device that can be used by individuals or in group settings, such as classrooms or community centers.

II. SYSTEM AND STRUCTURE

A. Electronics

- **MPU-6050** : It is a small, low-power, 6-axis MEMS (micro-electromechanical system) sensor module that combines a 3-axis gyroscope and a 3-axis accelerometer in a single package.
- **Flex Sensor**: It is a type of sensor designed to detect and measure the amount of bending or deflection in a material. It typically consists of a thin, flexible strip of material coated with a conductive material, such as carbon or metal. When the sensor is bent, the resistance of the material changes, as shown in Fig.2.

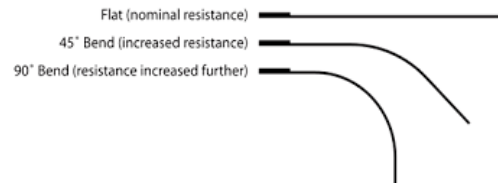


Fig. 2: Working of Flex Sensor

- **Arduino Nano 33 IoT**: It is a small development board based on the popular Atmel SAMD21 microcontroller, which is a robust low power Arm® Cortex®-M0 32-bit SAMD21 processor.

- **Arduino Nano:** The Arduino Nano is a small, versatile, easy-to-use development board based on the ATmega328P microcontroller.
- **Raspberry Pi 4 Model B (8GB):** It is a small single-board computer.
- **LED Matrix Display:** It is a display panel comprising an array of individual light-emitting diodes (LEDs) arranged in a grid or matrix formation. Each LED can be turned on or off independently, allowing for the creation of complex patterns, shapes, and animations.

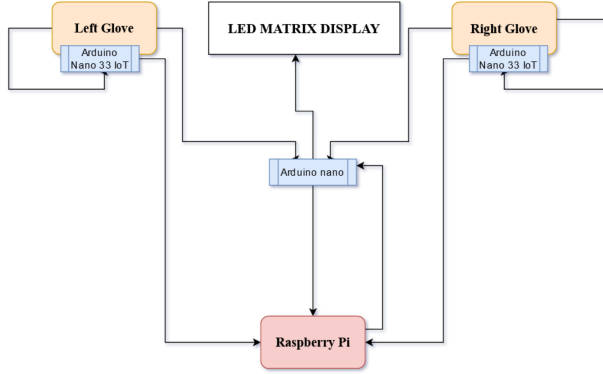


Fig. 3: Flowchart of Working of Electronics

B. The Machine Learning Model

1) *Data Collection:* The dataset consists of 14000 examples, with 7 different labels - Hello, G, E, S, T, R and Bye. There are 2000 examples of each label. A mean filter was applied to the data which took care of noisy oscillations in the readings and smoothened them.

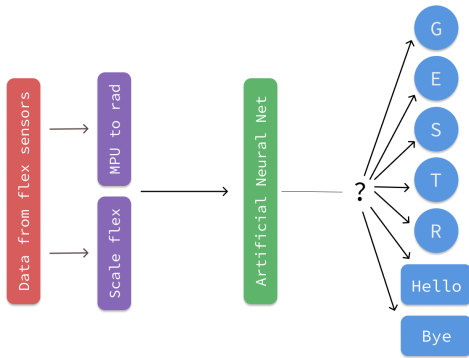


Fig. 4: Flowchart of machine learning model

2) Preprocessing:

- Each flex sensor has a different range of values, so standard scaling was applied on the flex sensors' readings to convert it into a standard normal distribution.
- The MPU readings are in degrees, at a much different scale than the standard scaled flex readings.

$$x_i^{scaled} = \frac{x_i - \mu_x}{\sigma_x}$$

We convert the MPU readings to radians to get values from $-\pi$ to π .

- Processing the features to get them to similar scales helps gradient descent converge much faster.
- The data was split into training and test datasets, with a test size of 30% and training size of 70%

3) Model Architecture:

- The input layer has 20 neurons
- There are 3 hidden layers of sizes 32, 64, 32 with ReLU activation function.
- All the layers had L2 regularization applied to get sparse weights and generalize the model.
- Finally the output layer has 7 neurons, with softmax activation applied to them.

$$y_i^{softmax} = \frac{e^{y_i}}{\sum_{i=0}^{n_{out}} e^{y_i}}$$

4) *Model Evaluation:* The model gained an accuracy of 91% on training data, while the accuracy on the test data was 88%

III. DESIGN

As shown in the figure, we have used eighteen 2.2" flex sensors on both hands, with nine on each hand, to sense the bend of each finger and wrist. There is one flex sensor on the thumb and little finger. There are two flex sensors on each index, ring, and middle finger, out of which one sensor is placed on the finger and the other is placed on the knuckles.

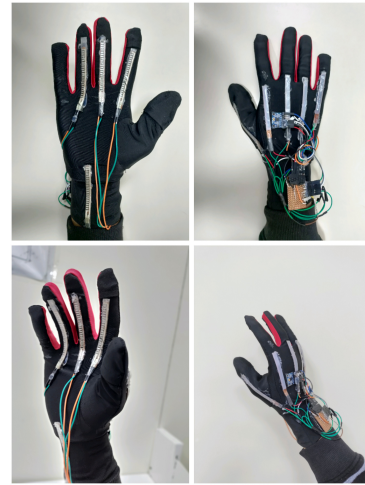


Fig. 5: Sensors Mounted Glove

We have also used MPU-6050 on the back of each hand, which gives orientation and movement with the help of a gyroscope and accelerometer. The six flex sensors on each hand and MPU-6050 are connected to the Arduino Nano 33 IoT attached to the glove, and the remaining three are attached to the Arduino Nano.

Arduino Nano 33 IoT and Arduino Nano are connected to the Raspberry Pi 4 Model B via a USB cable. While the Arduino Nano 33 IoT is attached to the glove, the Arduino Nano is



Fig. 6: Complete Design

attached to the shirt near the LED matrix display. The LED matrix display is placed on the abdomen. The Arduino Nano is further connected to the LED matrix display, which will display the translated text. The entire system is powered by a 10,000 mAh powerbank.

IV. WORKING PRINCIPLE

The flex sensors and MPU-6050 give analog input signals to Arduino Nano 33 IoT and Arduino Nano attached to the hands and chest, respectively. Serial communication between Arduino Nano 33 IoT and Arduino Nano and Raspberry Pi 4 Model B is established.

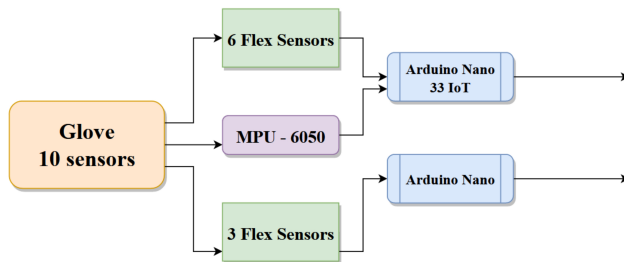


Fig. 7: Working of Glove

The analog signals are filtered using the Mean filter on Raspberry Pi Model B and given as input to the machine learning model for prediction. The machine learning model running in Raspberry Pi 4 Model B is an artificial neural network with 20 inputs and 7 output neurons. The output predicted by the model is displayed on the LED matrix display.

V. CONCLUSIONS

In conclusion, the development of a prototype to change sign language into translated text is a significant achievement in the field of assistive technology for the deaf and hard-of-hearing community. The prototype has shown promising results in accurately recognizing and translating sign language

gestures into text using machine learning algorithms.

The potential benefits of this technology are numerous, from enabling better communication and understanding between the hearing and deaf communities to improving educational and employment opportunities for the deaf and hard-of-hearing individuals. Moreover, the prototype has the potential to break down language barriers and facilitate social inclusion, empowering the deaf and hard-of-hearing community to participate actively in a wider range of activities.

However, it is essential to note that the prototype is still in its early stages of development, and there is still room for improvement. Further research and development are necessary to refine the technology's accuracy and usability and ensure that it is accessible and user-friendly for all individuals who rely on it.

Overall, the prototype for changing sign language into translated text holds great promise for advancing the field of assistive technology and has the potential to significantly enhance the quality of life and opportunities for the deaf and hard-of-hearing community.

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